

NEO Mathematics Learner Guide

Lesson 3 - Finding a Shorter Side

Driving question: How can I use the hypotenuse and one shorter side to find the missing shorter side?

What this lesson is about

In the previous lesson, the unknown side was the hypotenuse. This time, the hypotenuse is already known. Your job is to find one of the shorter sides. That means the route changes: instead of adding the two shorter-side squares, you subtract the known shorter-side square from the hypotenuse square.

The route

- Find the right angle.
- Trace to the side opposite it. That side is the hypotenuse.
- Identify the missing shorter side and its lower-case letter.
- Start with the theorem using the hypotenuse square on the left.
- Rearrange so the missing side square is alone.
- Subtract the known shorter-side square from the hypotenuse square.
- Square-root the result to return from a square area to a length.

Important idea

When the hypotenuse is missing, the method involves adding two square areas. When a shorter side is missing, the method involves subtracting a square area. The difference comes from the role of the unknown side, not from the way the triangle is turned on the page.

Example structure

If the hypotenuse is **b = 13 cm** and the known shorter side is **c = 12 cm**, then the missing side **a** is found by:

$$b^2 = a^2 + c^2$$

$$a^2 = b^2 - c^2$$

$$a^2 = 13^2 - 12^2$$

$$a^2 = 169 - 144 = 25$$

$$a = \sqrt{25} = 5 \text{ cm}$$

Using the interactive

- Open one example at a time.
- Use the diagram first. Do not start by guessing whether to add or subtract.
- Complete the dropdowns in order.
- Use Focus this step when you want the Socratic Prompt and Hint to support a particular step.
- Use the Scratchpad to copy one route or explain why subtraction is needed.

Final evidence

For your final task, create one full shorter-side solution. Show the right angle, the hypotenuse, the missing shorter side, the rearranged equation, the subtraction and the square-root step.